

EQ-1xx Series Centrifugal Pump -- Mechanical Seal Assembly

(Excerpt)

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Applicable assets: EQ-101, EQ-102, EQ-103 (Rajgarh Petrochemical Complex, Production Unit 2)
Authority Level: 3 (OEM Manual / Approved Technical Specification)

1. Scope

This excerpt covers the mechanical seal assembly for the EQ-1xx series single-stage centrifugal pump, as installed on EQ-101, EQ-102, and EQ-103 at the customer's Production Unit 2 feed pump train.

2. Seal Assembly Specification

Current part number	FSL-2240B (effective 2025-01-15, per Service Bulletin FP-SB-2025-04)
Superseded part number	FSL-2240A (do not use for new installations after transition period)
Seal type	Balanced mechanical seal, single cartridge
Max continuous operating temperature	95 degC at seal faces
Max differential pressure across seal	12 bar

3. Maintenance Interval

Recommended seal inspection every 18 months, or immediately following any thermal excursion event exceeding the rated operating temperature for a cumulative duration greater than 4 hours.

4. Known Failure Modes

- Thermal cycling induced face wear: repeated start/stop or load-cycling operation causes uneven thermal expansion across seal faces, accelerating wear beyond the rated service life.
- Recommended mitigation: monitor bearing housing temperature trend via installed instrumentation; sustained oscillation with rising baseline is an early indicator and should trigger inspection ahead of the scheduled interval.

5. Reference

See Service Bulletin FP-SB-2025-04 for the current approved seal part number. Field service reports informing this revision are on file with Fischer Pumps Ltd. Technical Support.